



Government of Karnataka
DEPARTMENT OF TECHNICAL EDUCATION

Program	CSC/EC/EE/IT/MT	Semester	I/II
Course Name	Computer Aided Engineering Graphics (CAEG)	Type of Course	Practical
Course Code	24ME02P	Contact Hours	6 hours/week (78 hours/semester)
Teaching Scheme	L: T:P- 0:2:4	Credits	4
CIE Marks	25	SEE Marks	100

1. Rationale: Engineering Drawing is universal & effective language of engineers which strengthens the technological structure. It helps in communicating design ideas and technical information to engineers and other professionals throughout the design process. The objective of Engineering Drawing & CAD is to introduce the students, the techniques of drawing and also to visualize and represent of 3D objects in 2D planes.

2. Course Outcomes: At the end of the Course, the student will be able to:

CO-01	Execute and dimension the drawing for the given scale in CAD
CO-02	Draw three principal views for Points and Planes in CAD
CO-03	Draw Orthographic views and Isometric Views in CAD
CO-04	Create Solid model for the given Drawing in CAD

1.Course Content (Practical with instructional details)

WEEK	CO	PO	Tutorial (Use Sketch Book) (1+1=2 Hours per week)	Practical (CAD) (2+2=4 Hours per week)
1	1	1,4,7	Fundamentals of Engineering Drawing: - Introduction to Engineering Drawing - Need for Engineering Drawing - Instruments Used in Engineering Drawing - Types of Lines and Applications	- Familiarization of CAD window Commands like New file, Saving the file, Opening an existing drawing file, Undo, Redo, move commands, Menu bar, Tool bar, Task bar & Ribbon bar. - Practice basic CAD Line commands - Practice different types of Lines in CAD

2	1	1,4,7	Dimensioning: Introduction to dimensioning, Need for dimensioning & Elements of dimensioning	Practice CAD commands like arc, circle, square, rectangle, chamfer, Trim, Inclined lines, Extend, Extend to Next, Shell, Fillet, Group, Array and Mirror commands
			Dimensioning common features: Line, Radius, Diameter, Arc, Chord, Angles, Sphere, Chamfer, through hole, Blind hole, Counter bore & Counter Sink	- Dimension the above features in CAD - Ref Suggestive Questions under the heading Fundamental of Engineering Drawing
3	1	1,4,7	- Concept of Scale: Full size, reduced size & Enlarged size - System of dimensioning: Aligned system & Uni-direction system - (Ref Suggestive Questions under the heading Dimensioning)	- Execute Problems taught in tutorial session on Aligned system & Uni-direction system of dimensioning - Ref Suggestive Questions under the heading Dimensioning
			- Types of Dimensioning 1.Chain Dimensioning 2.Parallel Dimensioning (Ref Suggestive Questions under the heading Dimensioning)	- Execute Problems taught in tutorial session on Chain and Parallel Dimensioning - Ref Suggestive Questions under the heading Dimensioning
4	1,2	1,4,7	3.Combined Dimensioning 4. Progressive Dimensioning (Ref Suggestive Questions under the heading Dimensioning)	- Execute Problems taught in tutorial session on Combined Dimensioning and Progressive Dimensioning - Ref Suggestive Questions under the heading Dimensioning
			Introduction to Projections: - Plane of Projections and Views - Horizontal plane, Vertical Plane, Profile Plane, Front View, Top View, Side View - First angle & Third angle projection with symbolic representation	Draw the symbolic representation of First angle & Third angle projection
5	2	1,4,7	Projection of Points (in all four Quadrants) (Ref Suggestive Questions under the heading Projection of Points)	- Execute Problems taught in tutorial session on Projection of Points - Ref Suggestive Questions under the heading Projection of Points
			Projection of Points (in all four Quadrants) (Ref Suggestive Questions under the heading Projection of Points)	- Execute Problems taught in tutorial session on Projection of Points - Ref Suggestive Questions under the heading Projection of Points

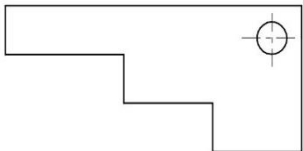
6	2	1,4,7	Projection of Planes • Projections of Planes (Triangular, Square, Pentagonal & Hexagonal lamina) for following conditions ➤ Lamina resting with one of its base Edge on HP ➤ Lamina resting with one of its Corner on HP (Ref Suggestive Questions under the heading Projection of Planes)	• Construct regular polygon: triangle, square, pentagon & hexagon • Execute Problems taught in tutorial session on Projection of Planes • Ref Suggestive Questions under the heading Projection of Planes
			• Projections of Planes (Triangular, Square, Pentagonal & Hexagonal lamina) for following conditions ➤ Lamina Lamina resting with one of its base Edge on HP ➤ Lamina resting with one of its Corner on HP (Ref Suggestive Questions under the heading Projection of Planes)	• Execute Problems taught in tutorial session on Projection of Planes • Ref Suggestive Questions under the heading Projection of Planes
7	3	1,4,7	Orthographic Projections Draw Orthographic Views for Pictorial drawings (Ref Suggestive Questions under the heading Pictorial Drawings)	• Execute Problems taught in the tutorial session on Orthographic Projection • Ref Suggestive Questions under the heading Pictorial Drawings
			Draw Orthographic projections for the Pictorial drawing (Ref Suggestive Questions under the heading Pictorial Drawings)	• Execute Problems taught in the tutorial session on Orthographic Projection • Ref Suggestive Questions under the heading Pictorial Drawings
8	3	1,4,7	Draw Orthographic projections for the Pictorial drawing (Ref Suggestive Questions under the heading Pictorial Drawings)	• Execute Problems taught in tutorial session on Orthographic Projection • Ref Suggestive Questions under the heading Pictorial Drawings
			Draw Orthographic projections for the Pictorial drawing (Ref Suggestive Questions under the heading Pictorial Drawings)	• Execute Problems taught in tutorial session on Orthographic Projection • Ref Suggestive Questions under the heading Pictorial Drawings
9	3	1,4,7	Isometric Projections Draw Isometric Projection for orthographic views (Ref Suggestive Questions under the heading Isometric Projections)	• Execute Problems taught in tutorial session on Isometric Drawing • Ref Suggestive Questions under the heading Isometric Projections
			Draw Isometric Projection for orthographic Views (Ref Suggestive Questions under the heading Isometric Projections)	• Execute Problems taught in tutorial session on Isometric Projection • Ref Suggestive Questions under the heading Isometric Projections

10	3		Draw Isometric Projection for orthographic views (Ref Suggestive Questions under the heading Isometric Projections)	- Execute Problems taught in tutorial session on Isometric Projection - Ref Suggestive Questions under the heading Isometric Projections
		1,4,7	Draw Isometric Projection for orthographic views (Ref Suggestive Questions under the heading Isometric Projections)	- Execute Problems taught in tutorial session on Isometric Projections - Ref Suggestive Questions under the heading Isometric Projections
11	4	1,4,7	Solid Modelling (Ref Suggestive Questions under the heading Orthographic and Pictorial Drawings)	Practice commands for Solid Modelling
			Solid Modelling (Ref Suggestive Questions under the heading Solid Modelling)	Create Solid Model
12	4	1,4,7	Solid Modelling (Ref Suggestive Questions under the heading Solid Modelling)	Create Solid Model
			Solid Modelling (Ref Suggestive Questions under the heading Solid Modelling)	Create Solid Model
13	4	1,4,7	Solid Modelling (Ref Suggestive Questions under the heading Solid Modelling)	Create Solid Model
			Solid Modelling (Ref Suggestive Questions under the heading Solid Modelling)	Create Solid Model

4. CIE and SEE Assessment Methodologies.

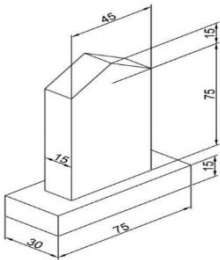
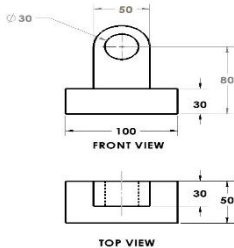
Sl. No	Assessment	Test Week	Duration (minutes)	Max marks	Average of all CIE=25 Marks
1.	CIE-1 Practice Test	5	180	25	
2.	CIE-2 Practice Test	9	180	25	
3	CIE-3 Practice Test	13	180	25	
Total Continuous Internal Evaluation (CIE)					25 Marks
Semester End Examination (SEE) -Practice			180	100	100 marks
Total Marks					125 Marks
Minimum marks to pass in CIE & SEE is 40% individually					

5. CIE 1 Practice Test Paper

Program	CSC/EC/EE/IT/MT			Semester	I/II
CourseName	Computer Aided Engineering Graphics (CAEG)			Test	CIE 1
Course Code	24ME02P	Duration	180	Marks	25
Name of the Course Coordinator:					
Questions				CO	Marks
1. Dimension the given drawing by adopting an Aligned system with Chain Dimensioning in CAD 				1	15
2. Draw three principal views of a point P , 30mm Above HP, 50mm behind VP & 40mm from Left Profile Plane.				2	10
Scheme of assessment for Q1		Scheme of assessment for Q2			
1. Execution – 10 Marks 2. Dimensioning- 05 Marks		1. Execution – 10 Marks			

Signature of the Course Coordinator
6. SEE- Model Practice Question Paper

Signature of the HOD

Program	Mechanical Engineering (General)		Semester	I/II
CourseName	Computer Aided Engineering Graphics	Course Code: 24ME02P	Duration	3 Hrs
Questions			CO	Marks
1. Draw orthographic views for the given Pictorial drawing. (only in CAD)			3	40
			4	60
2. Create 3D models for the given orthographic views (only in CAD)				
				
Scheme of assessment for Q1 1. Drawing three views-20 Marks 2. Adopting Drawing standards (Different types of Lines)-10 Marks 3. Dimensioning- 10 Marks		Scheme of assessment for Q2 1. Creating Solid model-50 Marks 2. List of commands used -10 Marks		
			Total Marks	100

Equipment/software list with Specification for a batch of 30 students

Sl. No.	Particulars	Specification	Quantity
01	CAD software	-	30 users

Suggestive Questions

Only for reference (Any other similar question shall be considered for CAD Practice, CIE and SEE)

1 Fundamentals of Engg. Drawing

- Draw Fig. 1 & Fig. 2 as per the drawing and identify types of lines.

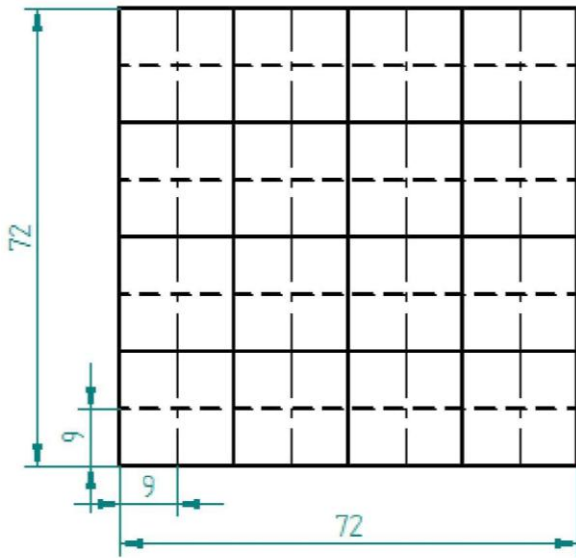


Fig. 1

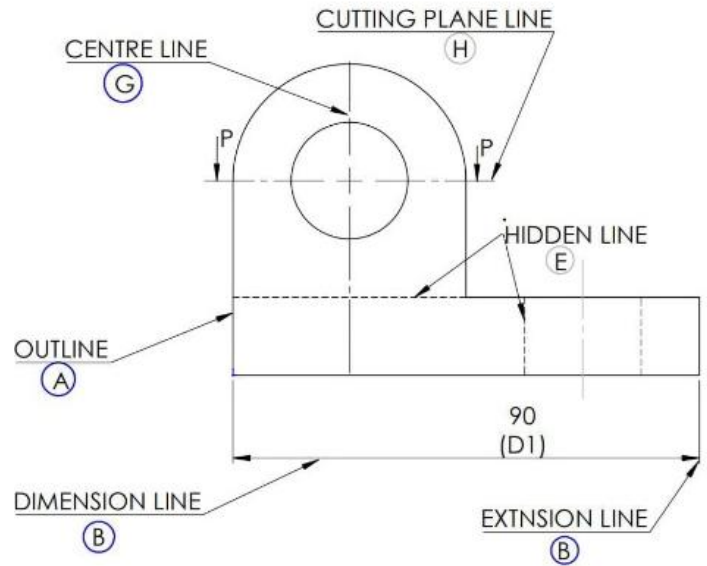


Fig. 2

2 Dimensioning

- Draw Fig. 3 to 1:1 scale, 1:2 scale & 2:1 scale.

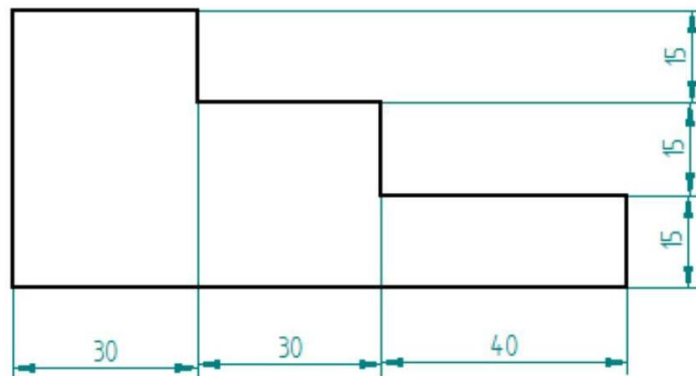


Fig. 3

- Copy Fig. 3 to 1:1 scale and dimension it using both Aligned system & Uni-directional system.
- Copy Fig. 4 to 2:1 Scale and dimension it using Aligned system with Chain dimensioning.

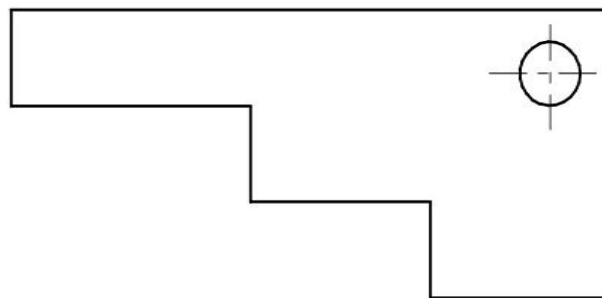


Fig. 4

- Copy Fig. 5 to 1:1 Scale and dimension it using Unidirectional system with Parallel dimensioning

- Copy Fig. 6 to 1:1 scale and dimension it using unidirectional system with Combined dimensioning method

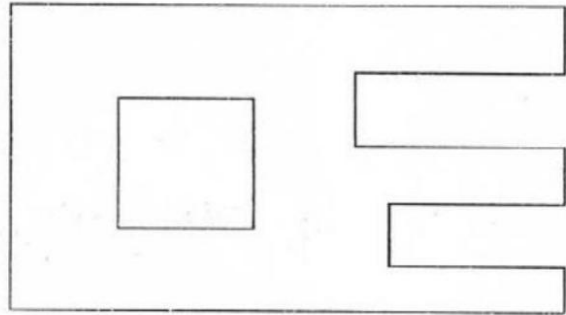


Fig. 6

- Copy Fig. 7 to 1:1 scale and dimension it using Aligned system with Progressive dimensioning method

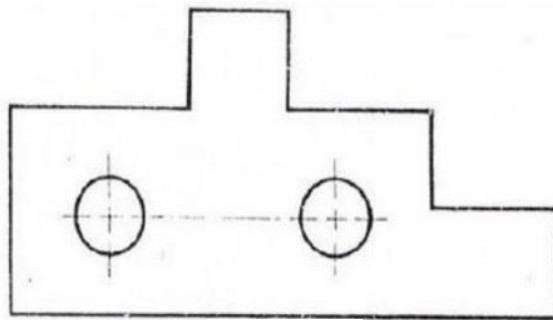


Fig. 7

- Copy Fig. 8 to 1:1 scale and dimension it using Aligned system with chain dimensioning method

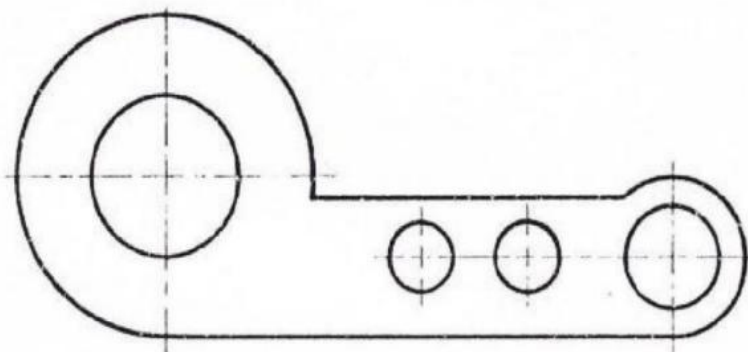


Fig. 8

3 Projection of Points

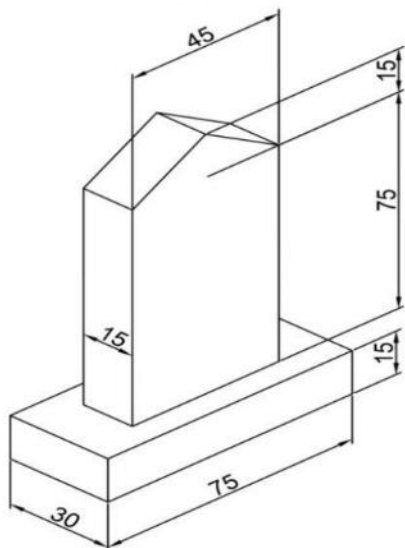
- Q1.** Draw three principal views of a point **P**, 30mm Above HP, 50mm in front of VP & 40mm from Left Profile Plane.
- Q2.** Draw three principal views of a point **P**, 30mm Above HP, 50mm behind VP & 40mm from Left Profile Plane.
- Q3.** Draw three principal views of a point **P**, 30mm Below HP, 50mm behind VP & 40mm from Left Profile Plane.
- Q4.** Draw three principal views of a point **P**, 30mm Below HP, 50mm in front VP & 40mm from Left Profile Plane.

4 Projections of Planes

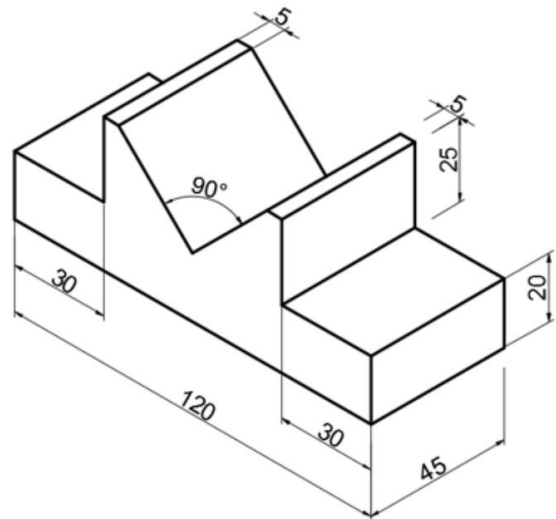
- Q1.** A triangular lamina of base edge 40mm rests with one its base edge on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q2.** A triangular lamina of base edge 40mm rests with one its Corner on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q3.** A square lamina of base edge 40mm rests with one its base edge on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q4.** A square lamina of base edge 40mm rests with one its corner on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q5.** A pentagonal lamina of base edge 40mm rests with one its base edge on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q6.** A pentagonal lamina of base edge 40mm rests with one its corner on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q7.** A hexagonal lamina of base edge 40mm rests with one its base edge on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q8.** A hexagonal lamina of base edge 40mm rests with one its corner on HP so that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.
- Q9.** A circular lamina of 30mm diameter rests on HP such that the surface of the lamina is inclined at 30° to HP. Draw the projections of the lamina.

5 Orthographic Projections(Pictorial Drawings)

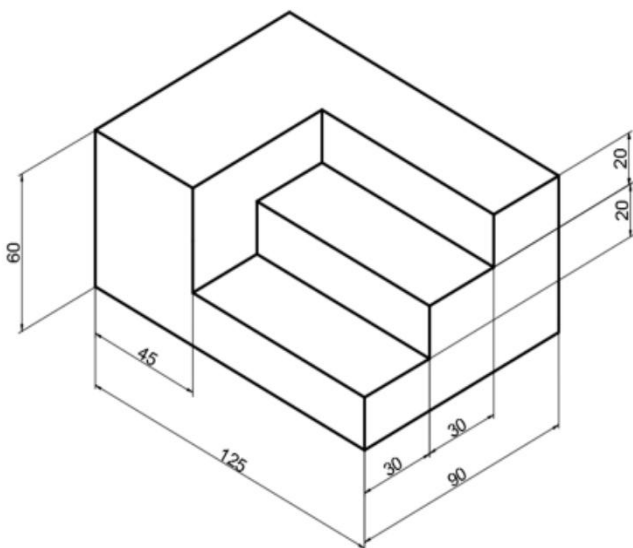
Q1. Draw the three principal views of the given component.



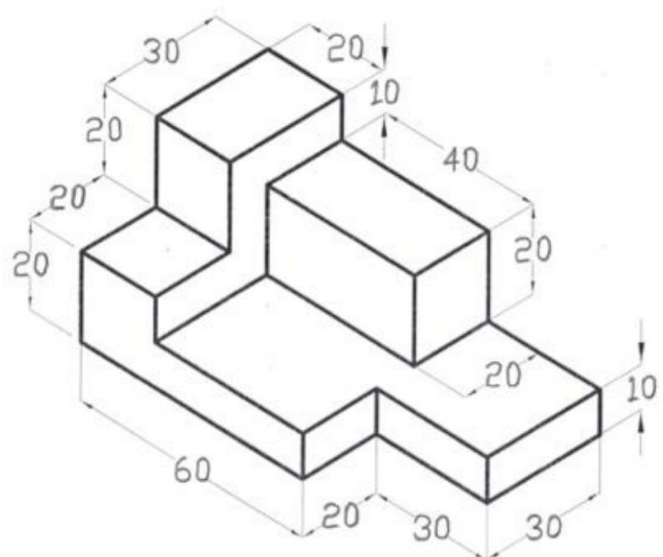
Q2. Draw the three principal views of the given component.



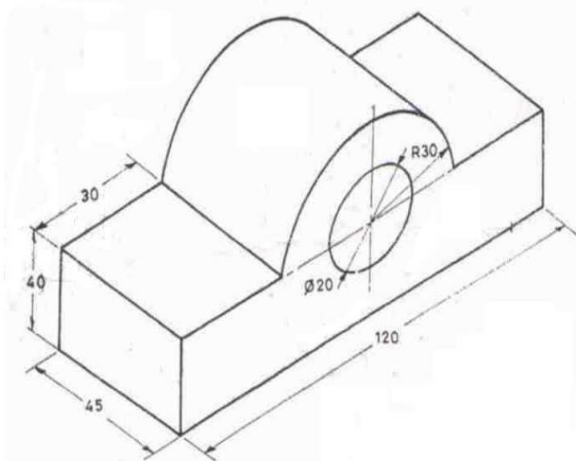
3. Draw the three principal views of the given component.



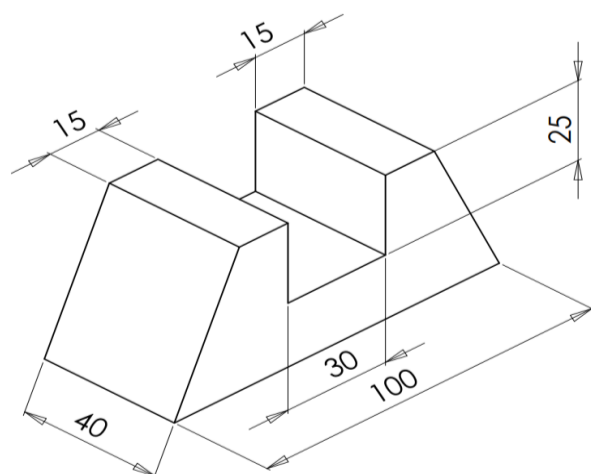
4. Draw the three principal views of the given component.



Q5. Draw the three principal views of the given component.



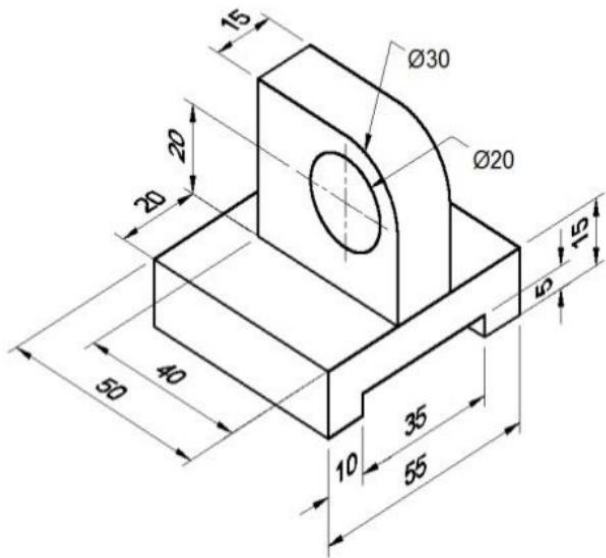
Q6. Draw the three principal views of the given component.



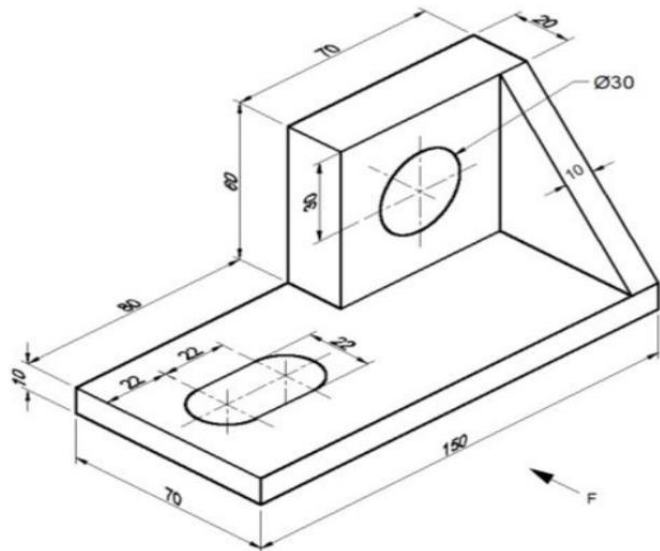
7. Draw the three principal views of the given

8. Draw the three principal views of the given

component.



component.



6. Isometric Projections

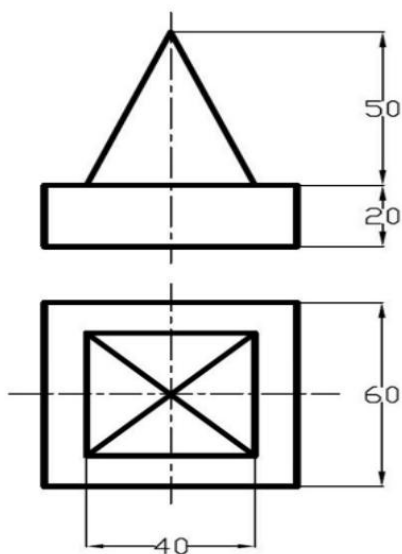
Q1. A cube of side 30 mm rests centrally on the top of another cube of side 60 mm. Draw the isometric view of the solid.

Q2. A square pyramid of base edge 50 mm and height 80 mm rests on the top of the cube of side 100 mm. Two sides of the base of the pyramid are parallel to the top edges of the cube. Draw the isometric view of the solid.

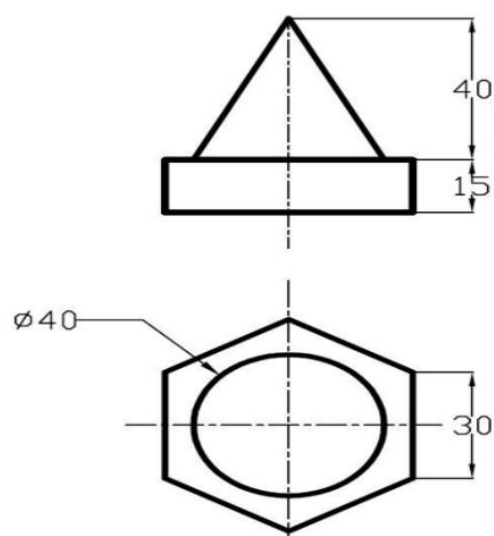
Q3. A cylindrical block of 40 mm diameter and length 50 mm is resting vertically on the centre of the cube of 70 mm side. Draw the isometric view of solids.

Q4. A square pyramid of base side 22 mm and height 20 mm rests centrally on the top of a cylinder of base diameter 40 mm and height 30 mm

Q5. Draw the isometric view of the machine component whose orthographic views are given below

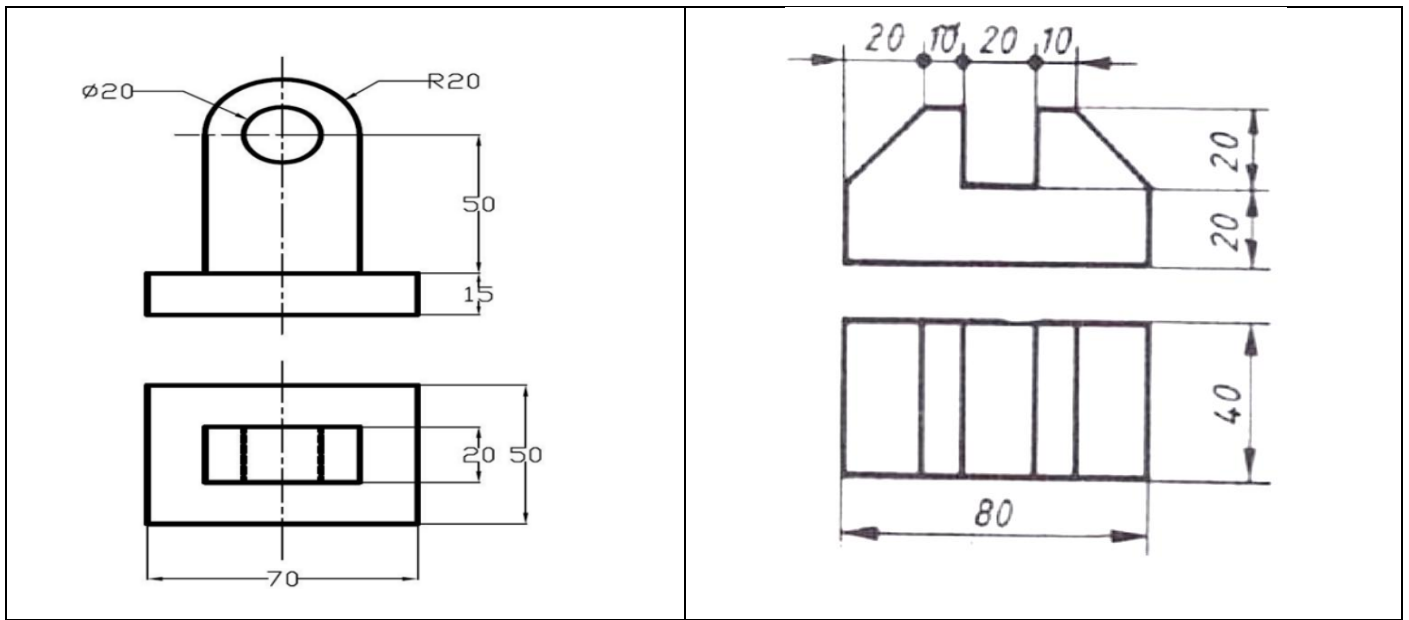


Q6. Draw the isometric view of the machine component whose orthographic views are given below

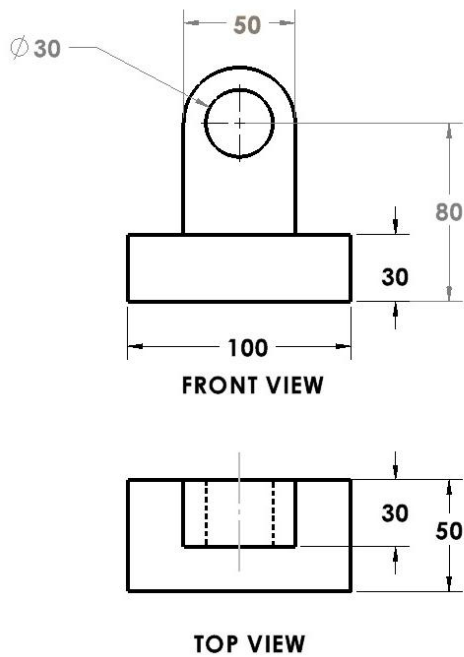


7. Draw the isometric view of the machine component whose orthographic views are given below

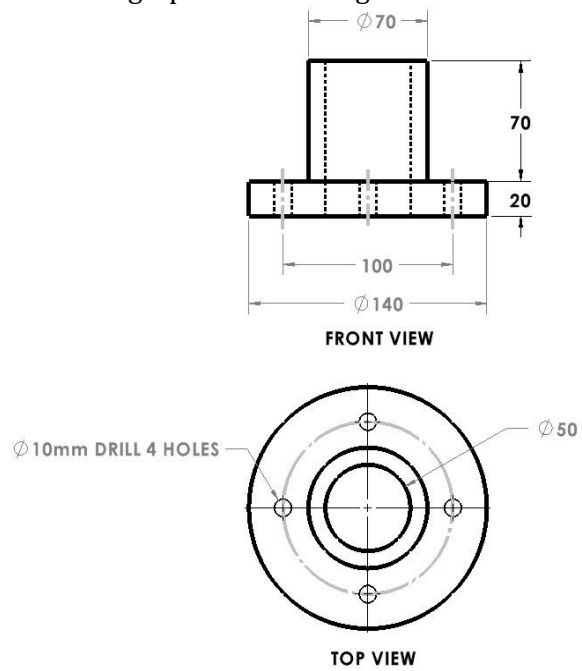
8. Draw the isometric view of the machine component whose orthographic views are given below



Q1. Create 3D model using CAD for the machine component whose orthographic views are given below

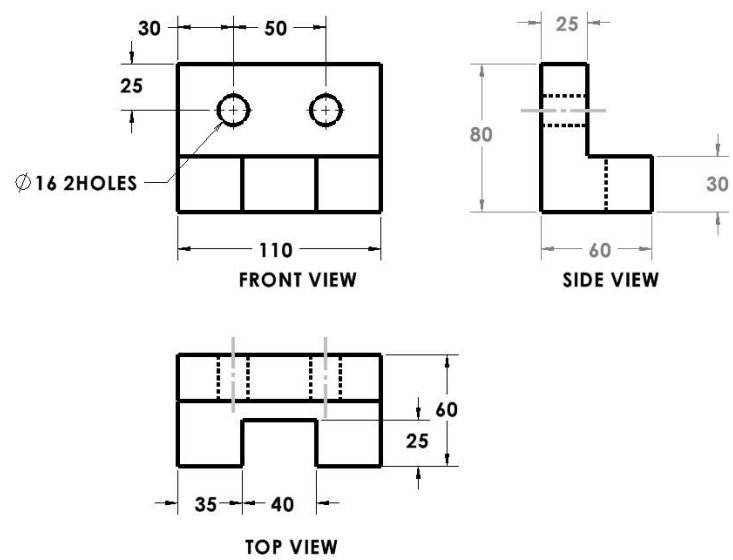
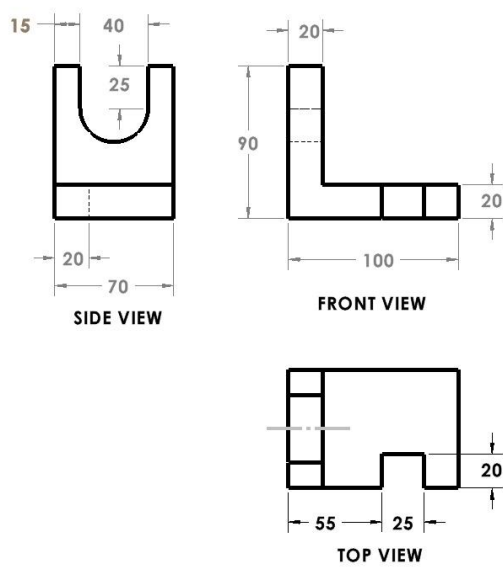


Q2. Create 3D model using CAD for the machine component whose orthographic views are given below



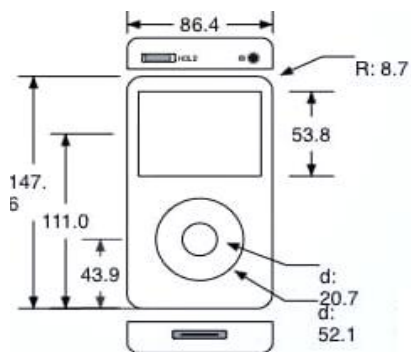
3. Create 3D model using CAD for the machine component whose orthographic views are given below

4. Create 3D model using CAD for the machine component whose orthographic views are given below

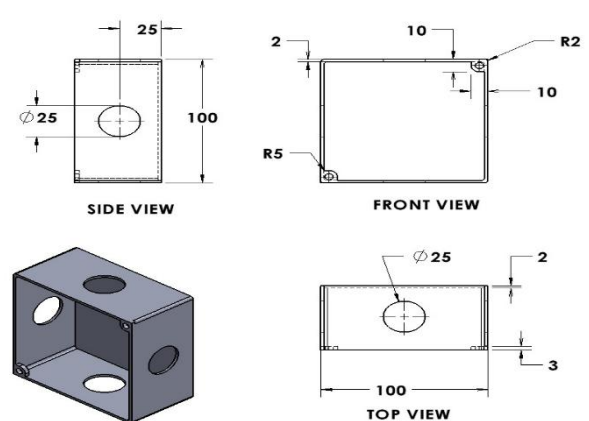


7. Solid Modelling

- 5 Create 3D model using CAD for the component whose orthographic views are given below



- 6 Create 3D model using CAD for the component whose orthographic views are given below



7 Create 3D model using CAD for the component whose orthographic views are given below

